Probability by counting

L2-CO2 Level 2 Intermediate · Combinatorics

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Turn probability questions into counting questions using equally likely outcomes, complements, tree diagrams and areas.
By the end you can set up equally likely outcomes and compute probabilities by counting; use complements, tree diagrams and the idea of independence without double counting; solve simple geometric probability problems by comparing areas.
Prerequisites: Counting without missing anything (L2-CO1). **Contests:** IMC, Grey Kangaroo, Pink Kangaroo, AMC 8, AMC 10. **Time:** about 75 minutes.

1 Lesson

Two fair dice are thrown. Which is more likely, a total of 11 or a total of 12? Leibniz, one of the inventors of calculus, once argued that they were equally likely, since 11 can only be made as $5+6$ and 12 only as $6+6$. He was wrong, and the reason is the main idea of this unit.

Paint one die red and the other blue. Now there are $6 \times 6 = 36$ outcomes (red score, blue score), and by symmetry they are all equally likely. A total of 12 happens only as $(6,6)$, but 11 happens as $(5,6)$ and as $(6,5)$. So the probability of 11 is $\frac{2}{36}$ and the probability of 12 is $\frac{1}{36}$, so an 11 is twice as likely. The mistake was to count outcomes that were not equally likely. Once the dice are distinguished, the problem is a counting problem.

1.1 Equally likely outcomes

If an experiment has N equally likely outcomes, and F of them make the event E happen, then

$$P(E) = \frac{F}{N}.$$

The formula needs the outcomes to be equally likely, and the way to ensure this is to make every object distinguishable. Number identical counters in a bag, and call two identical coins first and second. Labelling changes nothing physical, so labelling cannot change any probability, but labelling turns the outcomes into a product of choices that you can count with the methods of the previous unit.

You may choose whether order matters, as long as you are consistent. If you draw 2 counters from a bag of 5, you can count ordered pairs ($5 \times 4 = 20$ outcomes) or unordered pairs ($\binom{5}{2} = 10$ outcomes). Each unordered pair corresponds to exactly 2 ordered ones, so every probability comes out the same. What you must never do is count the favourable outcomes one way and the total the other.

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1.2 Complements and "at least one"

Exactly one of "E happens" and "E does not happen" occurs, so $P(\text{not } E) = 1 - P(E)$. Whenever a question says "at least one", look at the complement first. "At least one six in four rolls" needs several cases, but "no six in four rolls" has probability $\left(\frac{5}{6}\right)^4 = \frac{625}{1296}$, so the answer is $1 - \frac{625}{1296} = \frac{671}{1296}$.

1.3 Stages, independence and tree diagrams

Many experiments happen in stages. If the first stage gives event A with probability p, and then the second stage gives B with probability q (worked out knowing that A happened), then "A and then B" has probability pq. This is the multiplication principle again, since a fraction q of a fraction p of all the equally likely outcomes. When the second probability does not depend on the first stage at all, as with separate dice or coins, the stages are **independent**. Drawing without replacement is not independent, because the second draw depends on what has already gone.

Tree diagrams.
Multiply along a branch to get the probability of that whole path. Add the probabilities of the different paths that give the event you want. The paths cannot happen together, so adding is safe.

A tree is a picture of the multiplication and addition principles. Its advantage is that the branch probabilities need not be equal, so a tree copes with biased coins and players of different strengths, where a list of equally likely outcomes would be clumsy.

1.4 Geometric probability

Sometimes a number is chosen "uniformly at random" from an interval, or a point from a region. Then there are infinitely many outcomes and counting is impossible. Instead of counting, we measure length or area.

If a point is chosen uniformly at random in a region of area A, the probability that the point lies in a part of area F is $\frac{F}{A}$. If two numbers x and y are chosen independently and uniformly from intervals, the pair (x, y) is a uniform point in a rectangle, so any question about x and y becomes a question about areas.

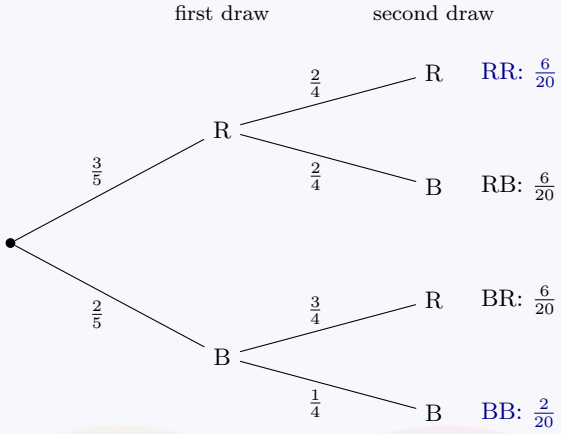
"Uniformly" means that equal areas get equal probability, which is the continuous version of "equally likely". The method is almost always the same. Draw the rectangle, draw the boundary lines of the condition, shade, and find the area, usually by subtracting simple triangles.

Example 1.
A bag contains 3 red counters and 2 blue counters. Two are taken out at random, without replacement. What is the probability that they are the same colour?

Solution.
A wrong argument says the outcomes are "two red", "one of each" and "two blue", so the answer is $\frac{2}{3}$. Those three outcomes are not equally likely, so this argument is invalid.
Number the counters R_1, R_2, R_3, B_1, B_2. There are $\binom{5}{2} = 10$ equally likely pairs. Two reds can be chosen in $\binom{3}{2} = 3$ ways and two blues in $\binom{2}{2} = 1$ way, so the probability is $\frac{4}{10} = \frac{2}{5}$.

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A tree diagram gets the same answer. The first counter is red with probability $\frac{3}{5}$, and then the second is red with probability $\frac{2}{4}$, since 2 of the 4 remaining counters are red. Two blues has probability $\frac{2}{5} \times \frac{1}{4}$. Adding the two paths gives $\frac{6}{20} + \frac{2}{20} = \frac{2}{5}$.



Use whichever method is quicker, and the other as a check. Note also that the tree shows "one of each" has probability $\frac{6}{20} + \frac{6}{20} = \frac{3}{5}$, which is far from the $\frac{1}{3}$ the wrong list suggested.

Example 2.
Four fair dice are rolled. What is the probability that at least two of them show the same number?

Solution.
"At least two the same" could mean one pair, two pairs, three of a kind or four of a kind. Casework is possible but slow. The complement is a single event, namely that all four dice are different.
Label the dice first to fourth, giving $6^4 = 1296$ equally likely outcomes. For all different there are 6 options for the first die, 5 for the second, 4 for the third and 3 for the fourth, so $6 \times 5 \times 4 \times 3 = 360$ outcomes. The probability that all are different is $\frac{360}{1296} = \frac{5}{18}$, so the answer is $1 - \frac{5}{18} = \frac{13}{18}$.
Note that with only four dice a repeat is much more likely than not. This is the same effect as in the birthday problem.

Example 3.
Asha and Ben play a best-of-three match, where the first to win two games wins the match. Asha wins each game with probability $\frac{2}{3}$, independently of the other games. What is the probability that Asha wins the match?

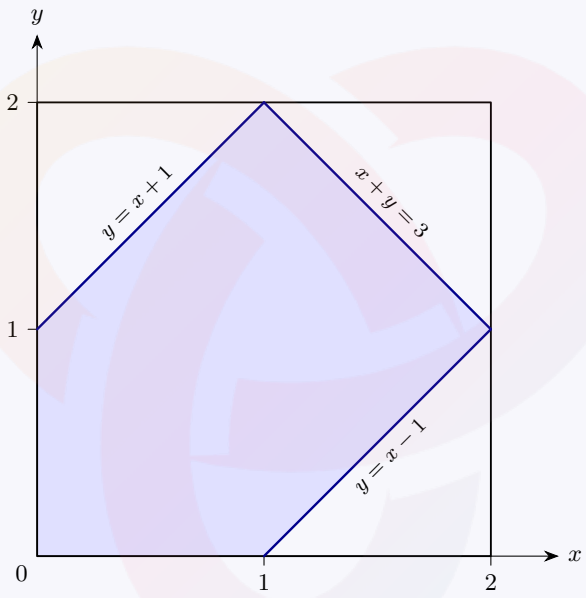
Solution.
The awkward feature is that the match might stop after two games or go to three, so the possible outcomes have different lengths. A tree diagram deals with this. Writing the winner of each game in turn, Asha wins the match as AA, ABA or BAA. These have probabilities $\frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$, and $\frac{2}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{27}$ for each of the other two, giving $\frac{12}{27} + \frac{8}{27} = \frac{20}{27}$.
Alternatively, suppose the players always play all three games, even when the match is already decided. That changes nothing about who wins the match, since a player with two wins keeps them. Now Asha wins the match exactly when she wins at least two of the three

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games. She wins all three with probability $\left(\frac{2}{3}\right)^3 = \frac{8}{27}$, and exactly two with probability $3 \times \left(\frac{2}{3}\right)^2 \times \frac{1}{3} = \frac{12}{27}$, where the 3 counts which game she loses. Again the answer is $\frac{20}{27}$.
Extending an experiment to a fixed length, so that every outcome has the same length, is worth remembering. This replaces a stopping rule by a count.

Example 4.
Real numbers x and y are chosen independently and uniformly at random between 0 and 2. What is the probability that $|x - y| < 1$ and $x + y < 3$?

Solution.
Two numbers chosen uniformly means a point (x, y) chosen uniformly in the square $0 \le x \le 2$, $0 \le y \le 2$, which has area 4. So draw the square and both conditions before calculating anything.



The condition $|x - y| < 1$ says that the point lies between the lines $y = x + 1$ and $y = x - 1$. These cut off two corner triangles, at the top left and the bottom right, each with legs 1 and area $\frac{1}{2}$, so the band has area $4 - 1 = 3$. The condition $x + y < 3$ removes the corner triangle with vertices $(1, 2)$, $(2, 1)$ and $(2, 2)$, which also has area $\frac{1}{2}$.
Before subtracting, check that this triangle really lies inside the band, or we would be removing area that was never there. Every point in the triangle has $1 \le x \le 2$ and $1 \le y \le 2$, so $|x - y| \le 1$. The only exceptions are boundary points, which have zero area and never matter. So the favourable region has area $3 - \frac{1}{2} = \frac{5}{2}$, and the probability is $\frac{5/2}{4} = \frac{5}{8}$.
This overlap check is the step most scripts omit. Whenever you subtract several pieces, ask whether they overlap each other and whether they lie inside the region you started with.

Common traps.

- Listing outcomes that are not equally likely, such as dice totals, or "two red, one of each, two blue".
- Mixing ordered and unordered counting in the same fraction.

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- Adding the probabilities of events that can happen together. The chance of at least one six in two rolls is not $\frac{1}{6}+\frac{1}{6}$, because a double six is counted twice. The correct value is $1-\frac{25}{36}=\frac{11}{36}$.
- Multiplying probabilities as if the stages were independent when drawing without replacement.
- Following only one branch of a tree. One red and one blue happens as RB *or* BR.
- In geometric problems, letting the region spill outside the square, or subtracting two pieces that overlap.

1.5 How to spot this in a contest

The words "at random", "fair" and "probability" tell you the topic, and the first step is to translate into counting. Ask the following questions in order. What are the equally likely outcomes, and how many are there? Is the event or its complement easier to count? Does the experiment happen in stages with unequal probabilities, in which case a tree is better than a list?

For anything chosen "uniformly" from an interval, draw a square immediately. On the IMC and the AMC 10 a probability answer can also be checked, since the answer must lie between 0 and 1, symmetric situations must give equal answers, and an event and its complement must add to 1. A rough estimate (is this nearer $\frac{1}{4}$ or $\frac{3}{4}$?) often eliminates two or three of the five options before you finish the calculation, and this check catches arithmetic slips.

2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

Problem 1 $\star$ Two fair six-sided dice are rolled. What is the probability that the product of the two scores is even?

Problem 2 $\star\star$ A drawer holds 4 black socks and 6 white socks. Three socks are taken out at random. What is the probability that all three are the same colour?

Problem 3 $\star\star$ A point is chosen uniformly at random inside a square of side 4. What is the probability that the point is more than 1 unit away from every vertex of the square?

Problem 4 $\star\star\star$ The five letters of the word MATHS are arranged in a random order, with all 120 orders equally likely. What is the probability that A is not at either end and M is somewhere to the left of S?

Problem 5 $\star\star\star$ Priya and Quentin play a match that ends as soon as one of them has won 3 games. Priya wins each game with probability $\frac{2}{3}$, independently of the other games, and there are no draws. What is the probability that the match lasts exactly 4 games?

Problem 6 $\star\star\star\star$ A stick of length 1 is broken at two points chosen independently and uniformly at random along its length, giving three pieces. What is the probability that every piece is shorter than $\frac{2}{3}$?

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3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 8 2016 Problem 13** ⋆
   Count the pairs. Which pairs give the required product?
2. **AMC 8 2011 Problem 18** ⋆
   36 equally likely ordered outcomes. Use symmetry between the two rolls
3. **AMC 10A 2003 Problem 12** ⋆⋆
   Draw the rectangle and the line. Compare areas
4. **AMC 10B 2017 Problem 9** ⋆⋆
   Add the cases of exactly two right and all three right
5. **UKMT IMC 2023 Question 20** ⋆⋆
   List only the outcomes that satisfy the given information
6. **AMC 10B 2018 Problem 6** ⋆⋆
   What must the first two chips look like?
7. **UKMT Pink Kangaroo 2020 Question 16** ⋆⋆⋆
   Divisible by 18 means divisible by 2 and by 9. Check the digit sum first
8. **AMC 8 2016 Problem 21** ⋆⋆⋆
   Imagine drawing every chip. What does the last chip tell you?
9. **UKMT Pink Kangaroo 2025 Question 16** ⋆⋆⋆
   Tree diagram, so track which bricks are available at each step
10. **UKMT IMC 2026 Question 25** ⋆⋆⋆
    Count sets of days. Remember that the week wraps round
11. **AMC 10A 2020 Problem 16** ⋆⋆⋆
    By symmetry, look at a single unit square. Compare areas

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4 Hints

1. When is a product of two whole numbers odd?
2. Count the sets of three socks, treating the socks as labelled.
3. Near each corner, which points are within 1 unit of that corner? What shape do they form?
4. Place A first. Then think about pairing up arrangements by swapping M and S.
5. What must the score be after 3 games, and who must win the fourth game?
6. Can two of the pieces both have length at least $\frac{2}{3}$? Then work out, as an area, the probability that one particular piece is that long.

5 Solutions

Problem 1

Two fair six-sided dice are rolled. What is the probability that the product of the two scores is even?

A product of two whole numbers is odd only when both numbers are odd. Each die is odd with probability $\frac{1}{2}$, independently, so the product is odd with probability $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. The product is even with probability $1 - \frac{1}{4} = \frac{3}{4}$.

Answer: $\frac{3}{4}$

Problem 2

A drawer holds 4 black socks and 6 white socks. Three socks are taken out at random. What is the probability that all three are the same colour?

Label the socks so that all are different. There are $\binom{10}{3} = 120$ equally likely sets of three socks. The sets of three black socks number $\binom{4}{3} = 4$ and the sets of three white socks number $\binom{6}{3} = 20$. So the probability is $\frac{4+20}{120} = \frac{24}{120} = \frac{1}{5}$.

Answer: $\frac{1}{5}$

Problem 3

A point is chosen uniformly at random inside a square of side 4. What is the probability that the point is more than 1 unit away from every vertex of the square?

The points of the square within 1 unit of a given vertex form a quarter of a disc of radius 1, with area $\frac{\pi}{4}$. Since the side is 4, which is more than twice the radius, the four quarter discs do not overlap, so together they have area π. The square has area 16, so the probability of landing within 1 unit of some vertex is $\frac{\pi}{16}$, and the required probability is $1 - \frac{\pi}{16}$.

Answer: $1 - \frac{\pi}{16}$

Problem 4

The five letters of the word MATHS are arranged in a random order, with all 120 orders equally likely. What is the probability that A is not at either end and M is somewhere to the left of S?

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We count favourable arrangements. A must go in one of the 3 middle positions. The other four letters then fill the remaining four positions in $4! = 24$ ways. Swapping M and S turns an arrangement with M to the left of S into one with M to the right of S and back again, so these 24 arrangements split into pairs and exactly 12 of them have M to the left of S. There are therefore $3 \times 12 = 36$ favourable arrangements, and the probability is $\frac{36}{120} = \frac{3}{10}$.

Answer: $\frac{3}{10}$

Problem 5

Priya and Quentin play a match that ends as soon as one of them has won 3 games. Priya wins each game with probability $\frac{2}{3}$, independently of the other games, and there are no draws. What is the probability that the match lasts exactly 4 games?

After 3 games the score is either 3 to 0 or 2 to 1. If the score is 3 to 0 the match ended after 3 games. If the score is 2 to 1, the match lasts exactly 4 games when the player leading 2 to 1 wins the fourth game, and lasts 5 games otherwise.

Priya leads 2 to 1 after 3 games with probability $3 \times \left(\frac{2}{3}\right)^2 \times \frac{1}{3} = \frac{4}{9}$, where the 3 counts which of the first three games she lost. She then wins game 4 with probability $\frac{2}{3}$, giving $\frac{8}{27}$. Quentin leads 2 to 1 with probability $3 \times \left(\frac{1}{3}\right)^2 \times \frac{2}{3} = \frac{2}{9}$, and then wins game 4 with probability $\frac{1}{3}$, giving $\frac{2}{27}$. These cannot both happen, so the answer is $\frac{8}{27} + \frac{2}{27} = \frac{10}{27}$.

Answer: $\frac{10}{27}$

Problem 6

A stick of length 1 is broken at two points chosen independently and uniformly at random along its length, giving three pieces. What is the probability that every piece is shorter than $\frac{2}{3}$?

Let the break points be at distances x and y from the left end, so (x, y) is a uniform point in the unit square, of area 1. The pieces have lengths equal to the smaller of x and y, the difference $|x - y|$, and 1 minus the larger of x and y.

Two pieces cannot both have length at least $\frac{2}{3}$, since together they would be longer than the stick. So the event that some piece has length at least $\frac{2}{3}$ splits into three cases with no overlap, one for each piece, and we can add their probabilities.

The left piece has length at least $\frac{2}{3}$ when $x \geq \frac{2}{3}$ and $y \geq \frac{2}{3}$, a square of side $\frac{1}{3}$ with area $\frac{1}{9}$. The right piece has length at least $\frac{2}{3}$ when $x \leq \frac{1}{3}$ and $y \leq \frac{1}{3}$, again area $\frac{1}{9}$. The middle piece has length at least $\frac{2}{3}$ when $|x - y| \geq \frac{2}{3}$, which is two corner triangles with legs $\frac{1}{3}$, of total area $2 \times \frac{1}{18} = \frac{1}{9}$. So some piece has length at least $\frac{2}{3}$ with probability $\frac{1}{3}$, and every piece is shorter than $\frac{2}{3}$ with probability $1 - \frac{1}{3} = \frac{2}{3}$.

Answer: $\frac{2}{3}$

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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